



A-level MATHEMATICS 7357/3

Paper 3

Mark scheme

June 2025

Version: 1.0 Final



Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

No student should be disadvantaged on the basis of their gender identity and/or how they refer to the gender identity of others in their exam responses.

A consistent use of 'they/them' as a singular and pronouns beyond 'she/her' or 'he/him' will be credited in exam responses in line with existing mark scheme criteria.

Further copies of this mark scheme are available from [aqa.org.uk](https://www.aqa.org.uk)

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Mark scheme instructions to examiners

General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- marking instructions that indicate when marks should be awarded or withheld including the principle on which each mark is awarded. Information is included to help the examiner make his or her judgement and to delineate what is creditworthy from that not worthy of credit
- a typical solution. This response is one we expect to see frequently. However credit must be given on the basis of the marking instructions.

If a student uses a method which is not explicitly covered by the marking instructions the same principles of marking should be applied. Credit should be given to any valid methods. Examiners should seek advice from their senior examiner if in any doubt.

Key to mark types

M	mark is for method
R	mark is for reasoning
A	mark is dependent on M marks and is for accuracy
B	mark is independent of M marks and is for method and accuracy
E	mark is for explanation
F	follow through from previous incorrect result

Key to mark scheme abbreviations

CAO	correct answer only
CSO	correct solution only
FT	follow through from previous incorrect result
'their'	indicates that credit can be given from previous incorrect result
AWFW	anything which falls within
AWRT	anything which rounds to
ACF	any correct form
AG	answer given
SC	special case
OE	or equivalent
NMS	no method shown
ISW	ignore subsequent working
PI	possibly implied
sf	significant figure(s)
dp	decimal place(s)

AS/A-level Maths/Further Maths assessment objectives

AO		Description
AO1	AO1.1a	Select routine procedures
	AO1.1b	Correctly carry out routine procedures
	AO1.2	Accurately recall facts, terminology and definitions
AO2	AO2.1	Construct rigorous mathematical arguments (including proofs)
	AO2.2a	Make deductions
	AO2.2b	Make inferences
	AO2.3	Assess the validity of mathematical arguments
	AO2.4	Explain their reasoning
	AO2.5	Use mathematical language and notation correctly
AO3	AO3.1a	Translate problems in mathematical contexts into mathematical processes
	AO3.1b	Translate problems in non-mathematical contexts into mathematical processes
	AO3.2a	Interpret solutions to problems in their original context
	AO3.2b	Where appropriate, evaluate the accuracy and limitations of solutions to problems
	AO3.3	Translate situations in context into mathematical models
	AO3.4	Use mathematical models
	AO3.5a	Evaluate the outcomes of modelling in context
	AO3.5b	Recognise the limitations of models
	AO3.5c	Where appropriate, explain how to refine models

Examiners should consistently apply the following general marking principles:

No Method Shown

Where the question specifically requires a particular method to be used, we must usually see evidence of use of this method for any marks to be awarded.

Where the answer can be reasonably obtained without showing working and it is very unlikely that the correct answer can be obtained by using an incorrect method, we must award **full marks**. However, the obvious penalty to students showing no working is that incorrect answers, however close, earn **no marks**.

Where a question asks the student to state or write down a result, no method need be shown for full marks.

Where the permitted calculator has functions which reasonably allow the solution of the question directly, the correct answer without working earns **full marks**, unless it is given to less than the degree of accuracy accepted in the mark scheme, when it gains **no marks**.

Otherwise we require evidence of a correct method for any marks to be awarded.

Diagrams

Diagrams that have working on them should be treated like normal responses. If a diagram has been written on but the correct response is within the answer space, the work within the answer space should be marked. Working on diagrams that contradicts work within the answer space is not to be considered as choice but as working, and is not, therefore, penalised.

Work erased or crossed out

Erased or crossed out work that is still legible and has not been replaced should be marked. Erased or crossed out work that has been replaced can be ignored.

Choice

When a choice of answers and/or methods is given and the student has not clearly indicated which answer they want to be marked, mark positively, awarding marks for all of the student's best attempts. Withhold marks for final accuracy and conclusions if there are conflicting complete answers or when an incorrect solution (or part thereof) is referred to in the final answer.

Section A

Q	Marking instructions	AO	Marks	Typical solution
1	Circles the second answer	2.3	R1	$\frac{1}{\pi} \times \pi = 1$
Question 1 Total			1	

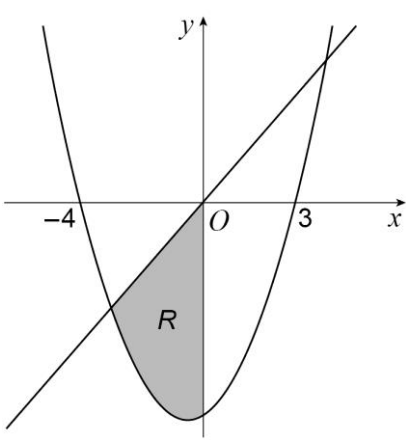
Q	Marking instructions	AO	Marks	Typical solution
2	Circles the third answer	1.2	B1	$\frac{\pi}{2}$
Question 2 Total			1	

Q	Marking instructions	AO	Marks	Typical solution
3	Ticks the fourth box	2.2a	R1	Increasing and convex
Question 3 Total			1	

Q	Marking instructions	AO	Marks	Typical solution
4(a)	Obtains 2	1.2	B1	2
Subtotal			1	

Q	Marking instructions	AO	Marks	Typical solution
4(b)	Uses at least one log rule correctly or substitutes a value of $a > 0$ correctly into the full expression leading to a positive value	1.1a	M1	$\log_a \sqrt{a} - \log_a \frac{1}{a^2} = \frac{1}{2} - -2$ $= \frac{5}{2}$
	Obtains 2.5 Do not ISW	1.1b	A1	
Subtotal			2	

Question 4 Total			3	
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Q	Marking instructions	AO	Marks	Typical solution
5	Shades one of regions I, II or III only or shades two of regions I, II or III only or shades regions I, II and III only	1.1a	M1	
	Shades correct region III only	1.1b	A1	
Question 5 Total			2	

Q	Marking instructions	AO	Marks	Typical solution
6(a)	Uses ${}^5C_1 \times 2^4 \times (-3x)$ or $2^5 \times 5 \times \left(-\frac{3}{2}x\right)$ or ${}^5C_2 \times 2^3 \times (-3x)^2$ or $2^5 \times 10 \times \left(-\frac{3}{2}x\right)^2$ with or without x PI by correct answer or $-240x$ or $720x^2$	1.1a	M1	$p = {}^5C_1 \times 2^4 \times (-3) = -240$ $q = {}^5C_2 \times 2^3 \times (-3)^2 = 720$
	Obtains $p = -240$ and $q = 720$ Allow if seen in... $-240x + 720x^2$... Condone $p = -240x$ and $q = 720x^2$	1.1b	A1	
Subtotal			2	

Q	Marking instructions	AO	Marks	Typical solution
6(b)	Substitutes $x = 0.02$ into $32 + px + qx^2 - 1080x^3$ with their p and q PI by correct answer Ignore any additional terms	2.2a	M1	$32 - 240 \times 0.02 + 720 \times 0.02^2 - 1080 \times 0.02^3 = 27.47936$
	Obtains 27.47936 CAO ISW	1.1b	A1	
Subtotal			2	

Question 6 Total			4	
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Q	Marking instructions	AO	Marks	Typical solution
7	Obtains $a + 5$, $2a + 5$ and $3a + 5$ PI by correct answer or $6a + 15$ ACF	2.5	B1	$a + 5 + 2a + 5 + 3a + 5 = 57$ $6a + 15 = 57$ $6a = 42$ $a = 7$
	Forms an equation using sum of three terms with at least two of $a + 5$, $2a + 5$ or $3a + 5$ or uses $S_n = \frac{n}{2}(2a + (n - 1)d)$ with at least two of $n = 3$, first term $a + 5$ and $d = a$ or uses $S_n = \frac{n}{2}(a + L)$ with at least two of $n = 3$, first term $a + 5$ and $L = 3a + 5$ or uses $6a + 15$ in an equation Must use 57 in the equation PI by correct answer	3.1a	M1	
	Obtains 7	1.1b	A1	
Question 7 Total			3	

Q	Marking instructions	AO	Marks	Typical solution
8(a)	Uses a suitable method and finds a value for A or B. eg rearranges and substitutes values or compares coefficients or uses cover-up method or uses inspection PI by $A = 1$ or $B = -1$	1.1a	M1	$x = A(2x+1) + B(x+1)$ $x = -1 \Rightarrow -1 = -A$ $A = 1$ $x = -\frac{1}{2} \Rightarrow -\frac{1}{2} = \frac{1}{2}B$ $B = -1$
	Obtains $A = 1$ Allow if seen in $\frac{1}{x+1}$	1.1b	A1	
	Obtains $B = -1$ Allow if seen in $\frac{-1}{2x+1}$	1.1b	A1	
	Subtotal		3	

Q	Marking instructions	AO	Marks	Typical solution
8(b)	Uses partial fractions with their A and B to obtain $A\ln(x+1)$ or $\frac{B}{2}\ln(2x+1)$ OE Condone B rather than $\frac{B}{2}$ Condone missing bracket or sign error of A or B	1.1a	M1	$\int_0^4 \frac{x}{2x^2+3x+1} dx = \int_0^4 \frac{1}{x+1} - \frac{1}{2x+1} dx$ $= \left[\ln(x+1) - \frac{1}{2}\ln(2x+1) \right]_0^4$ $= \ln 5 - \frac{1}{2}\ln 9 - \left(\ln 1 - \frac{1}{2}\ln 1 \right)$ $= \ln 5 - \ln 3$ $= \ln \frac{5}{3}$
	Obtains $\ln(x+1) - \frac{1}{2}\ln(2x+1)$ OE	1.1b	A1	
	Substitutes both limits correctly into their integrated expression of two $\ln$ terms (the subtraction does not need to be seen) Condone 0 for $\left(\ln 1 - \frac{1}{2}\ln 1 \right)$	1.1a	M1	
	Completes reasoned argument to obtain $\ln \frac{5}{3}$ Allow unsimplified fraction or correct recurring decimal for $\frac{5}{3}$ CSO	2.1	R1	
	Subtotal		4	
	Question 8 Total		7	

Q	Marking instructions	AO	Marks	Typical solution
9(a)	States one of the following transformations A Stretch in the y-direction, scale factor of 3 or B translation $\begin{bmatrix} 4 \\ 0 \end{bmatrix}$ or C stretch in the x-direction, scale factor of 3 or D translation $\begin{bmatrix} 4 \\ 3 \\ 0 \end{bmatrix}$ Only accept vectors for translations Allow 'parallel' or 'axis' for direction of stretch	3.1a	M1	Stretch in the y-direction by scale factor 3 and followed by translation $\begin{bmatrix} 4 \\ 0 \end{bmatrix}$
	Describes the sequence of two transformations in the correct order A followed by B or B followed by A or C followed by B or D followed by C Only accept vectors for translations Allow 'parallel' or 'axis' for direction of stretch	1.1b	A1	
	Subtotal		2	

Q	Marking instructions	AO	Marks	Typical solution
9(b)	States $x = 4$	1.1b	B1	$x = 4$
	Subtotal		1	

Q	Marking instructions	AO	Marks	Typical solution
9(c)	Explains that there could be more than one solution between the two points (3 and 5)	2.2b	E1	There could be more than one solution between the two points. The two points are either side of the asymptote.
	Explains that the two points (3 and 5) are either side of the asymptote or explains that it is discontinuous between the two points (3 and 5)	2.3	E1	
	Subtotal		2	

	Question 9 Total		5	
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Q	Marking instructions	AO	Marks	Typical solution
10(a)	Obtains $R = \sqrt{29}$ AWFW [5.38, 5.4]	1.1b	B1	$R = \sqrt{29}$
	Uses compound angle identity to obtain $R \sin \alpha = 5$ or $R \cos \alpha = 2$ or obtains $\tan \alpha = \frac{5}{2}$ or $\alpha = \text{AWFW}$ [68, 68.2] Allow angle in radians	1.1a	M1	$\tan \alpha = \frac{5}{2}$ $\alpha = 68.2$ Hence $\sqrt{29} \sin(x + 68.2)$
	Obtains $\sqrt{29} \sin(x + 68.2)$ For $\sqrt{29}$, AWFW [5.38, 5.4] and for 68.2, AWFW [68, 68.2]	2.1	R1	
Subtotal			3	

Q	Marking instructions	AO	Marks	Typical solution
10(b)(i)	Deduces $90 - \text{their } 68.2$ Ignore rounding after value of $90 - \text{their } 68.2$ seen	2.2a	B1F	$90 - 68.2 = 21.8$
Subtotal			1	

Q	Marking instructions	AO	Marks	Typical solution
10(b)(ii)	Deduces $14.2 + \text{their } \sqrt{29}$ Condone missing or wrong units Accept answer with more than 1 decimal place	2.2a	B1F	$14.2 + \sqrt{29} = 19.6^\circ \text{C}$
Subtotal			1	

Q	Marking instructions	AO	Marks	Typical solution
10(b)(iii)	Equates a complete model for the temperature to 15 eg $14.2 - \text{their} \sqrt{29} \sin(d + 68.2) = 15$ or $14.2 - (2 \sin d^\circ + 5 \cos d^\circ) = 15$ Allow use of inequality Condone use of another letter for d throughout PI by correct answer or angles	3.4	M1	$14.2 - \sqrt{29} \sin(d + 68.2) = 15$ $\sin(d + 68.2) = -\frac{0.8}{\sqrt{29}}$ $d + 68.2 = -8.54, 188.54, 351.46$ $d = 120.34, 283.26$ $\frac{283.26 - 120.34}{7} = 23 \text{ weeks}$
	Rearranges their equation correctly to obtain $\sin(d + \text{their} 68.2) = \dots$ Allow use of inequality PI by correct answer or angles	1.1a	M1	
	Obtains AWFW [188.2, 189] and AWFW [351, 351.5] or obtains AWFW [120, 120.8] and AWFW [282.8, 283.3] Ignore additional values PI by correct answer	1.1b	A1	
	Subtracts their two positive values of d or their two positive values of $d + \text{their} 68.2$ and divides by 7 PI by correct answer	3.1b	M1	
	Obtains AWFW [23.1, 23.4] or 23 ISW	1.1b	A1	
	Subtotal		5	

	Question 10 Total		10	
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Q	Marking instructions	AO	Marks	Typical solution
11(a)	Forms a correct expression for the surface area of the block of ice Can be unsimplified terms or states that the surface area is proportional to x^2	3.1b	B1	$S.A. = 2(8x^2 + 2x^2 + 4x^2) = 28x^2$ $\frac{dV}{dt} = -28kx^2$ $x = 4, \frac{dV}{dt} = -7$
	States $\frac{dV}{dt} = m(SA)$ or $\frac{dV}{dt} = -m(SA)$ or $\frac{dV}{dt} = m(\text{their } 28x^2)$ or $\frac{dV}{dt} = -m(\text{their } 28x^2)$ or $\frac{dV}{dt} = mx^2$ or $\frac{dV}{dt} = -mx^2$ Accept any letter for m	3.3	M1	$\Rightarrow -7 = -28k \times 16$ $\Rightarrow k = \frac{7}{28 \times 16} = \frac{1}{64}$ $\Rightarrow \frac{dV}{dt} = -\frac{28}{64}x^2 = -0.4375x^2$
	Substitutes $\frac{dV}{dt} = -7$ and SA = their '28' $\times 4^2$ into their differential equation $\frac{dV}{dt} = m(SA)$ or $\frac{dV}{dt} = -m(SA)$ or substitutes $\frac{dV}{dt} = -7$ and $x = 4$ into their differential equation of the form $\frac{dV}{dt} = m(28x^2)$ or $\frac{dV}{dt} = -m(28x^2)$ or $\frac{dV}{dt} = mx^2$ or $\frac{dV}{dt} = -mx^2$ Accept any letter for m	3.3	M1	
	Completes reasoned argument to show the given result with no incorrect expression for surface area and no incorrect substitution seen. AG To be awarded R1, marks M1M1 must be scored as the minimum Ignore units	2.1	R1	
Subtotal			4	

Q	Marking instructions	AO	Marks	Typical solution
11(b)	Forms a correct expression for the volume of the block of ice or states $8x^3$	3.1b	M1	$V = 4x \times 2x \times x$ $= 8x^3$ $\frac{dV}{dx} = 24x^2$
	Obtains $24x^2$	1.1b	A1	
Subtotal			2	

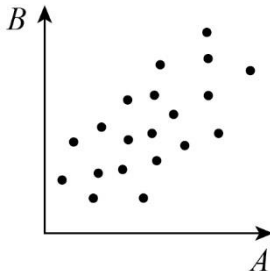
Q	Marking instructions	AO	Marks	Typical solution
11(c)(i)	Uses the chain rule to connect $\frac{dV}{dt} = -0.4375x^2$, their $\frac{dV}{dx} = 24x^2$ from part (b) and $\frac{dx}{dt}$	3.1a	M1	$\frac{dx}{dt} = \frac{dx}{dV} \times \frac{dV}{dt}$ $= \frac{1}{24x^2} \times -0.4375x^2$ $= -\frac{7}{384}$
	Obtains $-\frac{7}{384}$ AWFW $-[0.018, 0.0183]$	1.1b	A1	
Subtotal			2	

Q	Marking instructions	AO	Marks	Typical solution
11(c)(ii)	Identifies the height/width/length/edge/side/dimension/measurement is decreasing at a constant rate	3.5a	E1	The height is decreasing at a constant rate of 0.018 cm per minute.
	States it occurs at the rate of $\frac{7}{384}$ or $[0.018, 0.0183]$ or $[0.036, 0.0366]$ or $[0.072, 0.0732]$ cm per minute FT their constant $\frac{dx}{dt}$ from part c(i)	3.5a	E1F	
Subtotal			2	

Question 11 Total			10	
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Q	Marking instructions	AO	Marks	Typical solution
12	States a and b are rational	2.1	B1	Let a and b be rational
	Forms the product of two rational numbers $\frac{m}{n} \times \frac{p}{q}$	3.1a	M1	$a = \frac{m}{n}$ and $b = \frac{p}{q}$ $c = ab = \frac{m}{n} \times \frac{p}{q} = \frac{mp}{nq}$
	Concludes the argument fully correctly by stating that c is rational and contradicts c is irrational and if c is irrational at least one of a or b is irrational. To be awarded R1, marks B1M1 must be scored with their $\frac{mp}{nq}$ seen	2.1	R1	Hence c is rational which contradicts that c is irrational. Therefore, if c is irrational at least one of a or b is irrational.
Question 12 Total			3	

Section B

Q	Marking instructions	AO	Marks	Typical solution
13	Ticks the bottom right box	2.2b	B1	
Question 13 Total			1	

Q	Marking instructions	AO	Marks	Typical solution
14	Ticks the second box	1.1b	B1	$a = 6$ and $b = 12$
Question 14 Total			1	

Q	Marking instructions	AO	Marks	Typical solution
15(a)(i)	Obtains AWFW [0.00377,0.0038]	1.1b	B1	0.00378
Subtotal			1	

Q	Marking instructions	AO	Marks	Typical solution
15(a)(ii)	Obtains AWFW [0.616, 0.62]	1.1b	B1	0.617
Subtotal			1	

Q	Marking instructions	AO	Marks	Typical solution
15(a)(iii)	Finds $P(X \leq 8) = [0.953, 0.954]$ or $P(X \leq 9) = [0.982, 0.983]$ or $P(X > 9) = [0.017, 0.0174]$ or $P(X \geq 9) = [0.0467, 0.047]$ Condone incorrect or no labels	1.1a	M1	$P(X \geq 9) = 1 - 0.9532$ $= 0.0468$
	Obtains AWFW [0.0467, 0.047]	1.1b	A1	
Subtotal			2	

Q	Marking instructions	AO	Marks	Typical solution
15(b)	States one of the following assumptions in context - probability/proportion of a diner being vegan is constant or always 20% or fixed or the same or does not change - event of a diner is vegan is independent from one diner to the next - only two types of diners vegan or non-vegan Do not ignore incorrect statements about any of the above Must use 'vegan' and 'diner' Allow equivalent statements for non-vegan diners for the reference to probability or independence Do not allow statements which imply probabilities are independent Ignore statement which implies there is a fixed number of diners	3.5b	E1	The event of a diner is vegan is independent from one diner to the next.
	Subtotal		1	
	Question 15 Total		5	

Q	Marking instructions	AO	Marks	Typical solution
16(a)(i)	Adds the correct area of each rectangle using scale factor eg. $17x$ or states that there are 34 big (cm) squares or states that there are 850 small (2mm) squares PI by their ratio or $1\text{cm} = 20$ seen on the vertical scale or by correct answer	3.1a	M1	Total area of rectangles $= 2 \times 1 + 1 \times 2 + 0.5 \times 3 + 1.5 \times 7$ $+ 0.5 \times 2$ $= 17$ $340 \div 17 = 20$ Total number of individuals who travel between 3 km and 5 km $= 20 \times (0.5 \times 2 + 0.5 \times 3 + 1 \times 7)$ $= 20 \times 9.5$ $= 190$
	Calculates their ratio PI by eg $340 \div \text{their } 17$ or 20 or $340 \div \text{their } 34$ or 10 or $340 \div \text{their } 850$ or 0.4 or by correct answer May be seen in calculation PI by their ratio or $1\text{cm} = 20$ seen on the vertical scale without incorrect working	1.1a	M1	
	States one of $340 \div \text{their } 17 \times 9.5$ $340 \div \text{their } 34 \times 19$ $340 \div \text{their } 850 \times 475$ May be embedded in calculation PI by correct answer	3.1a	M1	
	Obtains 190	1.1b	A1	
	Subtotal		4	

Q	Marking instructions	AO	Marks	Typical solution
16(a)(ii)	States one of the following reasons - The actual distance for each person is unknown - The distances are grouped Distances are not evenly distributed	3.5b	E1	The actual distance for each person is unknown
	Subtotal		1	

Q	Marking instructions	AO	Marks	Typical solution
16(b)	States 'negative'	2.2b	B1	Negative
	Subtotal		1	

Q	Marking instructions	AO	Marks	Typical solution
16(c)	States one of the following reasons - widths/intervals of class/group are different - gives an accurate/better representation of the data	2.4	E1	Class widths are different.
	Subtotal		1	

	Question 16 Total		7	
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Q	Marking instructions	AO	Marks	Typical solution
17(a)	Obtains 3	1.1b	B1	Mean = 3
	Subtotal		1	

Q	Marking instructions	AO	Marks	Typical solution
17(b)	Obtains 1.5 Accept AWRT 1.7	1.1b	B1	Variance = 1.5
	Subtotal		1	

Q	Marking instructions	AO	Marks	Typical solution
17(c)	Calculates the mean or variance of the given model $B(30, 0.1)$ or forms a correct equation in one variable to find n or p for binomial model using their mean from part (a) and their variance from part (b)	3.1a	M1	Mean = 30×0.1 = 3 Variance = $30 \times 0.1 \times 0.9$ = 2.7 The mean is the same but the variance is different, so this is not a suitable model.
	Obtains the correct mean 3 and variance 2.7 or obtains the correct $n=6$ and $p=0.5$ for binomial model Accept $n=7$ and $p=0.43$ for binomial model if using variance 1.7	1.1b	A1	
	Infers that it is not suitable because of different variance or infers teacher's belief not supported because of different variance or infers that it is not suitable because of different n and p from $n = 30$ and $p = 0.1$ Must make a single overarching conclusion Must have obtained correct mean from part (a) and correct variance from part (b) To be awarded R1, marks M1A1 must be scored	2.2b	R1	
	Subtotal		3	

Q	Marking instructions	AO	Marks	Typical solution
17(d)(i)	Recalls 'opportunity' or 'convenience' or 'opportunistic' sampling method	1.2	B1	Opportunity sampling.
Subtotal			1	

Q	Marking instructions	AO	Marks	Typical solution
17(d)(ii)	States one of the following advantage of sampling method - easy/simple - quick/fast - cheap/not expensive OE Do not accept 'convenient'	3.5a	E1	Easy to do.
Subtotal			1	

Question 17 Total			7	
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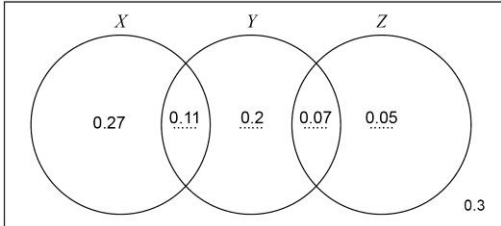
Q	Marking instructions	AO	Marks	Typical solution
18(a)	States the z value ($\pm$) 0.67(45) for the inverse normal distribution PI by [4.89, 4.9] or [6.5, 6.51] seen	3.4	M1	$a = 4.89$ $b = 6.51$ $IQR = 6.51 - 4.89$ $= 1.62 \text{ mmol/l}$
	Obtains correct lower quartile AWFW [4.89, 4.9] or obtains correct upper quartile AWFW [6.5, 6.51] CAO	1.1b	A1	
	Uses a complete method to find the IQR e.g. $b - a$ or $2 \times (5.7 - a)$ or $2 \times (b - 5.7)$ or states or calculates their b – their a	3.4	M1	
	Obtains AWFW [1.6, 1.62] Condone missing units	1.1b	A1	
Subtotal			4	

Q	Marking instructions	AO	Marks	Typical solution
18(b)	States $H_0 : \mu = 5.7$ $H_1 : \mu < 5.7$	2.5	B1	$H_0 : \mu = 5.7$ $H_1 : \mu < 5.7$
	States or uses correct model PI by normal with mean 5.7 and variance $\frac{1.2^2}{160}$ or 0.009 or standard deviation $\frac{1.2}{\sqrt{160}}$ or 0.095 or better OE or by correct probability AWFW [0.145, 0.147] or test statistic $(\pm) \frac{5.6 - 5.7}{1.2 \div \sqrt{160}}$ or test statistic value AWFW $(\pm)$ [1.05, 1.1] or critical value AWFW [5.54, 5.55]	1.1a	M1	$\bar{X} \sim N(5.7, \frac{1.2^2}{160})$ $P(\bar{X} < 5.6) = 0.146$ $0.146 > 0.05$ Do not reject H_0 There is insufficient evidence to suggest that the mean cholesterol level has reduced after taking the dietary supplement.
	Obtains AWFW [0.145, 0.147] or the correct test statistic value AWFW – [1.05, 1.1] or the correct critical value AWFW [5.54, 5.55] May be seen in expression for critical/acceptance regions	1.1b	A1	

Mark scheme continues on the next page

Q	Marking instructions	AO	Marks	Typical solution
18(b) cont.	Correctly compares their value of $P(< \text{ or } \leq 5.6)$ with 0.05 or correctly compares their negative test statistic with AWFW – [1.64, 1.65] or correctly compares 5.6 with their acceptance region or critical region or critical value May be seen on a diagram	3.5a	M1	
	Infers H_0 not rejected or H_0 accepted FT their comparison Condone reject H_1	2.2b	A1F	
	Concludes, from a fully correct comparison, in context by stating that there is insufficient evidence to suggest the mean cholesterol level has reduced after taking the dietary supplement. Conclusion must not be definite, eg use of ‘suggest’, ‘support’ etc To be awarded R1, marks M1A1M1A1 must be scored as the minimum	3.2a	R1	
	Subtotal		6	
	Question 18 Total		10	

Q	Marking instructions	AO	Marks	Typical solution
19(a)	Obtains 0.12	1.1a	B1	0.12
	Subtotal		1	

Q	Marking instructions	AO	Marks	Typical solution
19(b)	Obtains at least one of 0.11 or 0.2 or 0.07 or 0.05 in the correct place	1.1a	M1	
	Obtains at least two of 0.11 or 0.2 or 0.07 or 0.05 in the correct place	1.1a	M1	
	Completes the Venn diagram correctly	1.1b	A1	
	Subtotal		3	

Q	Marking instructions	AO	Marks	Typical solution
19(c)	States their 0.11 + their 0.07 PI by correct answer	1.1a	M1	$0.11 + 0.07 = 0.18$
	Obtains 0.18 FT their values provided the final positive answer is < 0.38	1.1b	A1F	
	Subtotal		2	

Q	Marking instructions	AO	Marks	Typical solution
19(d)	States $P(Y' Z') = \frac{P(Y' \cap Z')}{P(Z')}$ OE Condone missing $P(Y' Z')$ or calculates $P(Y' \cap Z')$ PI by 0.57 seen on numerator or calculates $1 - \text{their } P(Z)$ from part 19(a) PI by 0.88 seen on denominator PI by correct answer	1.1a	M1	$P(Y' Z') = \frac{P(Y' \cap Z')}{P(Z')}$ $= \frac{0.3 + 0.27}{1 - 0.12}$ $= \frac{0.57}{0.88}$ $= \frac{57}{88}$
	Obtains $\frac{57}{88}$ or AWFW [0.647, 0.65]	1.1b	A1	
	Subtotal		2	

Q	Marking instructions	AO	Marks	Typical solution
19(e)	States their $P(Z)$ from part 19(a) $\times 0.38$ or compares their $P(Y Z)$ with 0.38 or compares their $P(Z Y)$ with their $P(Z)$ from part 19(a) or any other valid comparison with one correct probability to at least 2 sf	1.1a	M1	$P(Y) \times P(Z) = 0.38 \times 0.12$ $= 0.0456$ $P(Y \cap Z) = 0.07$ $0.0456 \neq 0.07$ $P(Y \cap Z) \neq P(Y) \times P(Z)$ Hence events Y and Z are not independent.
	Completes a reasoned argument with correct values and concludes that Y and Z are not independent	2.1	R1	
	Subtotal		2	

	Question 19 Total		10	
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Q	Marking instructions	AO	Marks	Typical solution
20(a)	Obtains AWFW [0.387, 0.39]	1.1b	B1	0.3875
	Subtotal		1	

Q	Marking instructions	AO	Marks	Typical solution
20(b)	Obtains AWFW [0.84, 0.843]	1.1b	B1	0.8427
	Subtotal		1	

Q	Marking instructions	AO	Marks	Typical solution
20(c)(i)	Obtains AWFW [0.003, 0.0034]	1.1b	B1	0.0033
	Subtotal		1	

Q	Marking instructions	AO	Marks	Typical solution
20(c)(ii)	Deduces that $P(F = 0.6) = 0$	2.2a	E1	$P(F = 0.6) = 0$ since F is continuous.
	Explains the continuous nature of the normal distribution	2.4	E1	
	Subtotal		2	

Q	Marking instructions	AO	Marks	Typical solution
20(d)	Concludes correctly in context for the means Comparison must include the word such as ‘on average’ or ‘typically’ or ‘in general’ etc Must use ‘ oxides ’ or ‘ emissions ’ and ‘ 2000 ’ or ‘ 2010 ’ at least once throughout Ignore values	2.2b	E1	The oxides of nitrogen emissions by cars in 2000 are greater on average and less varied than those in 2010.
	Concludes correctly in context for the standard deviations Comparison must include the word such as ‘varies’, ‘spread’ ‘disperse’ ‘variation’ or ‘consistent’ etc Do not allow comparison that only includes ‘range’ or ‘variety’	2.2b	E1	
Subtotal			2	

Q	Marking instructions	AO	Marks	Typical solution
20(e)	States that there are some blanks in the LDS or states that there are some missing data in the LDS or states that the emissions are not known for every car in LDS	2.4	E1	There are some blanks on the oxides of nitrogen emissions in the LDS.
Subtotal			1	

Q	Marking instructions	AO	Marks	Typical solution
20(f)	States that there are no Nissan cars in the LDS or states that the LDS only has BMW, Ford, Toyota, Vauxhall and Volkswagen cars	2.4	E1	There are no Nissan cars in the LDS.
Subtotal			1	

Question 20 Total			9	
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Question Paper Total			100	
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